

4(a)	parental genotypes Aa × Aa and gametes A a and A a ; offspring genotypes AA Aa (Aa) aa ; offspring phenotypes (correctly linked to genotypes) normal normal / carrier (normal / carrier) OCA1A / albino / described ; probability 25% ; A 1 in 4 / 0.25 / ¼ I ratio	4
4(b)(i)	<i>missense</i> results in a different amino acid ; <i>nonsense</i> ref. to STOP codon ;	2
4(b)(ii)	any four from: 1 frameshift / described ; 2 change in, primary structure / amino acid sequence ; 3 protein / enzyme / tyrosinase, folds incorrectly / changes tertiary structure / changes 3D shape ; 4 may introduce STOP codon ; 5 shortened polypeptide / no tyrosinase produced ; 6 <u>tyrosinase</u>, non-functional / lacks active site / active site changed shape / cannot form ESCs ; 7 tyrosine not converted to DOPA / DOPA not converted to dopaquinone ; 8 (so) dopaquinone not formed to produce melanin ;	4

Question	Answer	Marks
4(c)	any two from: 1 small gene pool / low genetic variation / low genetic diversity / low heterozygosity / high homozygosity ; 2 isolated (population) / described ; 3 inbreeding (within their population) ; 4 albinism has a selective advantage ; 5 founder effect / genetic bottleneck ; 6 genetic drift ;	2